Homework 5 solutions

1. Since the two lists are sorted, they can be combined using `merge`(). Adding an element (including its extra links) takes constant time per link, since the lists are doubly-linked. The expected sizes of the two skip lists are $O(m)$ and $O(n)$, so the operation takes $O(n+m)$.

2. Let p be the probability of getting a head, and write $\alpha = \frac{1}{p}$. Recomputing $E[S_i]$ we have

$$\begin{aligned} E[S_i] &= \sum_{j=1}^{n} P(S_{i,j}=1) \\ &= \sum_{j=1}^{n} p^i \\ &= np^i \\ &= \frac{n}{\alpha^i} \end{aligned}$$

The expected search path length (horizontal moves) is

$$E[L] = \sum_{i=0}^{\log_\alpha n} E[L_i] + \sum_{i \geqslant 1+\log_\alpha n} E[L_i].$$

$E[L_i] = \alpha$, so the first summation is $\alpha(1+\log_a n)$. Also, $E[L_i] \leqslant E[S_i]$, so the second summation is bounded by

$$\begin{aligned} \sum_{i \geqslant 1+\log_\alpha n} E[L_i] &\leqslant \sum_{i \geqslant 1+\log_\alpha n} E[S_i] \\ &= \sum_{i \geqslant 1+\log_\alpha n} \frac{n}{\alpha^i} \\ &= \frac{n}{\alpha n} + \frac{n}{a^2 n} + \cdots + \\ &= \frac{2}{\alpha}. \end{aligned}$$

so

$$\begin{aligned} E[L] &\leqslant \alpha(1+\log_\alpha n) + \frac{2}{\alpha}. \\ &= \alpha \log_\alpha n + O(1) \\ &= \alpha \frac{\ln n}{\ln \alpha} + O(1). \end{aligned}$$

Taking derivative of $E[L]$ with respect to α, we have

$$\begin{aligned} (E[L])' &= (\ln n(\alpha \ln^{-1}\alpha))' \\ &= \ln n\left(\ln^{-1}\alpha - \alpha \ln^{-2}\alpha \frac{1}{\alpha}\right) \\ &= \ln n \ln^{-2}\alpha(\ln \alpha - 1). \end{aligned}$$

This expression is minimized when $\ln \alpha - 1 = 0$, or $\alpha = e$, or $p = e^{-1}$.

3.

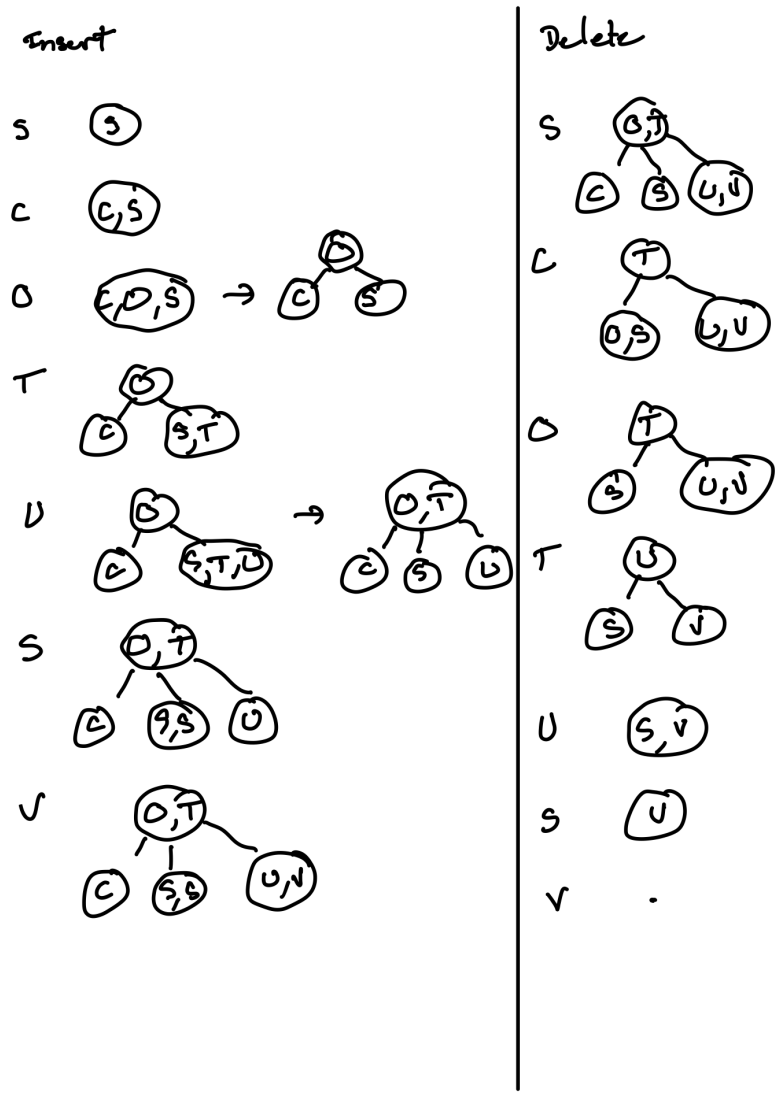

Figure 1.